

EDUCATION
BACKGROUND**University of Michigan at Ann Arbor**

M.S. in Robotics

Overall GPA: 4.0/4.0;

08/2024 - Present

University of North Carolina at Chapel Hill

B.S. in Applied Mathematics, B.S. in Physics

Overall GPA: 3.918/4.0;

GPA of Applied Mathematics: 3.89/4.0; GPA of Physics: 3.96/4.0

08/2020 - 05/2024

PUBLICATION
*DENOTES EQUAL
CONTRIBUTION

Song, J. *, **Ma, H. ***, Bagoren, O., Sethuraman, A. V., Zhang, Y., Skinner, K. A., “*OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework*,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. **Under Review**

Ma, H. *, Tamim, S. I., Guan, J. H., Sáenz, P. J., “*Solid Particles Walking on a Vibrating Interface*.” **Under Prep**

CONFERENCE
AND WORKSHOP
PRESENTATIONS“*OceanSim: A GPU-Accelerated Underwater Robot Perception Simulation Framework*”AQ²UASIM Workshop, *IEEE International Conference on Robotics And Automation (ICRA)*,

05/2025 Atlanta, GA

“*Solid Particles Walking on a Vibrating Interface*”Oral Presentation, *APS Division of Fluid Dynamics (DFD)*,

11/2024 Salt Lake City, UT

RESEARCH
PROJECT**Training Underwater Object Detection with Synthetic Data****Ann Arbor, MI**

Supervised by Prof. Katherine Skinner, Field Robotics Group (FRoG)

05/2025 - Present

- Built an object-based synthetic data generation pipeline in Isaac Sim, integrated with real-time underwater rendering effect for sim2real transfer.
- Leveraged SOTA imageTo3D generative network to fill the asset gap for replicating and improving an existing dataset with sim data generated from Omniverse.
- Extensive applications of openUSD SDK for scene authoring and shader coding for domain randomization.

Teleoperating KUKA Arm with Simulation-in-the-loop Digital Twin **Ann Arbor, MI**

Supervised by Prof. Dmitry Berenson, Auto Robotic Manipulation Lab (ARM) 05/2025 - 08/2025

- Developed ROS2 bridge for tele-operating Kuka in Isaac Sim environment with Apple Vision joystick and motion planning with MoveIt.
- Leveraged SDF toolkit for high fidelity simulation of forced manipulation task such as closing oil cap or screwing bolts.
- Used PhysX joint simulation for replicating deformable object behaviors such as bending due to gravity and fracture upon intense collision.
- The synthetic throughput can reach 30Hz output for both vision and force signal on a RTX 3090 machine.

Underwater Robots Digital Twin**Ann Arbor, MI**

Supervised by Prof. Katherine Skinner, Field Robotics Group (FRoG)

01/2025 - 05/2025

- 3D Reconstruction of Marine Hydrodynamics Laboratory Towing Tank (MHL, Department of Marine Engineering, UoM) from lidar scan and stereo camera.
- Created dynamic and high visual fidelity digital twin of underwater environments and BlueROV robot in Isaac Sim.
- Developed GPU-accelerated underwater sensor (Imaging/Side Scan Sonar, DVL, etc) through Nvidia Warp.

- Simulated realistic underwater column effects suitable to any water types with Underwater Image Formation Model and GPU rendering.

Solid Particles Walking on a Vibrating Interface

Chapel Hill, NC

Supervised by Pedro Sáenz, Physical Mathematics Lab (PML)

05/2023 - 05/2024

- Discovered period-doubling vertical dynamics and “walking” states for solid micro-particles $\sim 0.4\text{mm}$ diameter on a vibrated liquid-liquid interface.
- Characterized the “walking” regimes and speed by grayscale image processing and classic object tracking with Kalman Filter.
- Numerically computed and simulated particles’ hydrodynamical behaviors with self-developed PDE numerical solver.
- Designed and assembled the primary experiment structure for vibrational fluid dynamic research.
- Paper is expected to be submitted by 2025 Fall.

SCHOLARSHIP AND AWARDS

Daniel C. Johnson Outstanding Junior Award (Top 1 within department)

- Awarded annually to the physics major who is judged by the faculty to be the most outstanding student of the junior class. *Department of Physics and Astronomy, UNC* **04/2023**

Summer Undergraduate Research Fellowship (54 recipients, 2023)

- Granted to undergraduate students conducting research under a faculty mentor. \$4000 fellowship included. *Department of Undergraduate Research, UNC* **04/2023**

SKILLS

Software/Computing Literacy: Isaac Sim/Lab, Omniverse Kit, OpenUSD, ROS2, C/C++, CUDA, SolidWorks, Arduino, LabView, Unreal Engine, Illustrator, Premiere Pro, Adobe Substance Combo,

Experimental Techniques: High-Speed Videography, 3D Printing, Laser Cutting, Basic Electronics, Metal and Wood Crafting, NI-DAQ based Signal Processing, Particle Image Velocimetry

Professional Membership: *IEEE* and *APS* (American Physical Society)

Services: Reviewer of *IROS*

REFERENCES

Field Robotics Group (FRoG)

Prof. Katherine Skinner

Assistant Professor

Robotics, UoM

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Physical Mathematics Lab (PML)

Prof. Pedro Sáenz

Assistant Professor

Applied Mathematics, UNC

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